BLACK HOLE MICROSTATES OF TWO D6-BRANES CARRY NO ANGULAR MOMENTUM

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ABSTRACT. For the D2-D2-D2-D6 system on T^6 with any number of D6-branes, every classical supersymmetric vacuum is a simple quiver representation, so the vacua are the critical points of one function on a smooth space, and they come in pairs. With one D6-brane they are exactly 12 isolated points for generic moduli, and with two exactly 56 isolated points at explicit moduli, as Sen's zero angular momentum conjecture predicts. With three D6-branes we find all 208 predicted vacua, each isolated.

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1. INTRODUCTION

A supersymmetric black hole in four dimensions has a spherically symmetric horizon. Sen observed that the near-horizon geometry $AdS_2 \times S^2$ then forces every microstate of a single-centred black hole, once the fermion zero modes of the broken supersymmetries are removed, to be a singlet of the rotation group $SU(2)$ [22, 13].[1] In particular the protected index of such a black hole equals its degeneracy and is positive. The statement is about the black hole, where the string coupling is large. Chowdhury, Garavuso, Mondal and Sen proposed a stronger form: at a generic point of the moduli space of the compactification, even in the weakly coupled region where the microstates are described by D-branes, all the microstates of a single-centred black hole carry zero angular momentum [8, 9].

The strong form can be tested in type II string theory on T^6. There, for the charges carried by black holes, multi-centred configurations do not contribute to the index [20], and the index itself is known exactly from the $\mathcal{N} = 8$ counting formula [23, 21]. Chowdhury, Garavuso, Mondal and Sen studied N_1, N_2, N_3 D2-branes wrapped on the three two-tori T^2_{45}, T^2_{67}, T^2_{89} of $T^6 = T^2_{45} \times T^2_{67} \times T^2_{89}$ together with N D6-branes wrapped on all of T^6. With $N_1 = N_2 = N_3 = 1$ the low energy dynamics is a quantum mechanics with gauge group $U(1)^3 \times U(N)$ and four supercharges, the dimensional reduction of an $\mathcal{N} = 1$ gauge theory in four dimensions (Figure 1). The BPS states are the harmonic forms on the space $\mathcal{M}$ of classical vacua, with the rotation group acting as the Lefschetz $SU(2)$ [24, 14, 1]. The strong conjecture therefore says that $\mathcal{M}$ is a finite set of points, and its number of points must be the index

$$-\hat{c}(4N) = 12,\ 56,\ 208,\ 684,\ 2032,\ 5616, \ldots \qquad (N = 1, 2, 3, 4, 5, 6, \ldots), \tag{1}$$

where $\hat{c}(4N)$ is the coefficient of $q^N y^0$ in $-\vartheta_1(z|\tau)^2/\eta(\tau)^6$ (Appendix B).

In [8] the vacua were found for $N = 1$, and in [9] for $N = 2$ and $N = 3$: 12, 56 and 208 isolated solutions of the F-term equations in a fixed gauge, at random rational values of the parameters, with Hilbert series computed in Singular and Macaulay2 and numerical solutions

[1] The argument is that the near-horizon $SU(1,1|2)$ symmetry contains the rotation group, and that the black hole entropy counts states in the AdS_2 ground state, which is annihilated by all the generators of $SU(1,1|2)$, including $\vec{J}$. The relation between the index and the degeneracy that follows from it is derived in [13].

as a check; configurations where the gauge choice fails were examined separately and gave no solutions. For $N = 2$ the D-term equations and the gauge field potential were then analysed solution by solution. For $N = 3$ the authors state that they did not check explicitly that each solution of the F-term equations carries a unique solution of the D-term equations, and expected it on general grounds [9, Section 5]. Chowdhury and Maji returned to these equations with numerical algebraic geometry: they recovered 12, 56, 208 by homotopy continuation and Hilbert series, found 684 for $N = 4$, checked the D-term equations numerically for $N \leq 4$ [10], and found 2032 and 5616 for $N = 5, 6$ by monodromy [11]. These counts agree with (1), but they are numerical, and none of these works proves that the vacua are isolated or that their number is the predicted one.

In this paper we settle the structure for every N and the counts for $N \leq 2$, and for $N = 3$ we exhibit the predicted number of isolated vacua. Let $\mathcal{M}_c$ be the space of vacua, at values c of the parameters of the superpotential, modulo gauge transformations and modulo the flat directions of the centre of mass. Write $\mathcal{R}^{\mathrm{s}}$ for the space of simple representations of the quiver, $\mathrm{P}G$ for the gauge group modulo its centre and f_c for the function induced by the superpotential on the smooth variety $\mathcal{R}^{\mathrm{s}}/\mathrm{P}G$.

Theorem A. *Let $N \geq 1$ and let the parameters $c^{(k\ell)}$ of the superpotential be nonzero. Every solution of the F-term equations is a simple representation of the quiver. Consequently, for every value of the Fayet–Iliopoulos parameters:*

- *(i) every complex gauge orbit of F-term solutions contains exactly one unitary orbit of solutions of the D-term equations;*
- *(ii) at every vacuum the gauge fields vanish up to the overall translation, so the gauge field potential imposes nothing further;*
- *(iii) $\mathcal{M}_c$ is the critical locus of f_c on the smooth variety $\mathcal{R}^{\mathrm{s}}/\mathrm{P}G$, taken modulo the shift symmetries;*
- *(iv) the map $x \mapsto -x$ on all chiral fields induces an involution of $\mathcal{M}_c$ without fixed points. In particular, whenever $\mathcal{M}_c$ is finite, its number of points is even.*

Part (iv) explains why every number in (1) is even. Part (iii) turns the conjecture into a question about the critical points of one function on a smooth space, and we use it to give exact counts.

Theorem B.

- *(i) For $N = 1$ and all values of c outside a proper Zariski-closed subset, $\mathcal{M}_c$ consists of exactly 12 points.*
- *(ii) For $N = 2$ there are explicit rational parameters c_0 (Table 1) at which $\mathcal{M}_{c_0}$ consists of exactly 56 points, each a nondegenerate critical point of f_{c_0}. The count is exact: no vacuum is lost to the choice of gauge.*
- *(iii) For $N = 3$ and the same c_0, $\mathcal{M}_{c_0}$ contains 208 distinct vacua, each an isolated and nondegenerate critical point of f_{c_0}.*
- *(iv) For $N = 2, 3$ and every c close to c_0, $\mathcal{M}_c$ contains at least 56, respectively 208, nondegenerate vacua, and the same lower bound holds for all c outside a proper Zariski-closed subset.*

For $N = 1$ and $N = 2$ the vacua are isolated, so $P(y)$ is a constant and all BPS states have $J_3 = 0$: the strong zero angular momentum conjecture holds at these parameters, with the counts (1). For $N = 3$ the 208 vacua of part (iii) are isolated, their number is the index $-\hat{c}(12)$, and they carry $J_3 = 0$; what we cannot prove is that no further vacuum exists at c_0. We show that a further vacuum would need coordinates with the primes $10^9 + 7$ and $2^{30} + 3$ in their denominators (Proposition 5.1). Theorem A(i) supplies the step that was left open

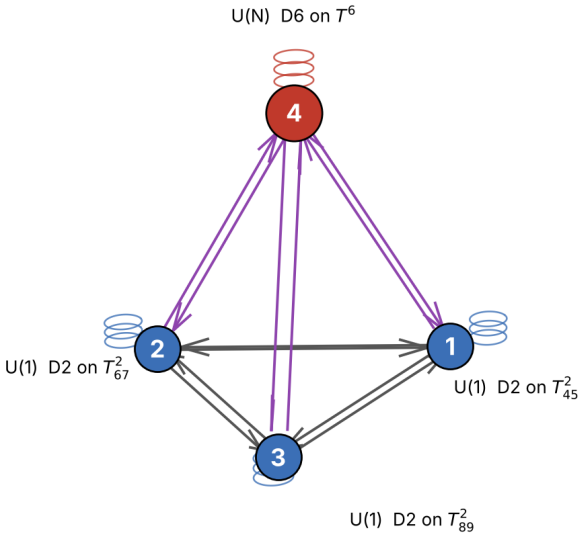


FIGURE 1. The quiver of the D2-D2-D2-D6 system drawn on a tetrahedron. Nodes $1, 2, 3$ carry $U(1)$ and node 4 carries $U(N)$. Each edge carries the two bifundamental fields $Z^{(k\ell)}$ and $Z^{(\ell k)}$, and each node carries three adjoint fields $\Phi_1^{(k)}, \Phi_2^{(k)}, \Phi_3^{(k)}$, drawn as loops.

for $N = 3$ in [9] and checked numerically in [10], and does so for every N. The statement for $N = 2$ at c_0 is a theorem: the number of solutions of the gauge-fixed equations is computed with exact rational arithmetic, it is matched by an explicit list of algebraic solutions, each of which is checked to satisfy every equation identically, and the configurations excluded by the gauge choice are shown to contain no solution.[2] For $N = 4$ we find exactly $684 = -\hat{c}(16)$ vacua in a gauge slice modulo a large prime (Section 5.5); this is not a proof.

The proof of Theorem A is short. The F-term equations for the adjoint fields say that the products of the bifundamentals around each edge are nonzero constants, and that on the $U(N)$ node the commutators of the adjoints are $-c^{(k4)}$ times the identity modulo a rank one matrix. A proper subrepresentation would make the commutator equation hold on a nonzero quotient or subspace with the rank one term removed, and taking the trace there gives $-c^{(k4)}$ times its dimension equal to zero. Simplicity then gives the D-terms by King's theorem [17], the gauge fields by Schur's lemma, and the pairing by the oddness of the superpotential.

Theorem B rests on a gauge slice that is valid at every simple representation where certain rows are independent, on exact elimination for the slice equations, and, for $N = 2$, on separate exact computations showing that the configurations where these rows are dependent carry no vacua. The rational univariate representation of the solutions [19] gives a certificate that can be

[2]The computations behind Theorem B are exact; no floating point number enters a proof. The programs are archived at [5].

checked independently of the program that produced it, and a second certificate, a determinant prime to the eliminating polynomial, shows that every solution found is nonsingular.

The paper is organised as follows. Section 2 recalls the quantum mechanics and states the conjecture precisely. Section 3 proves Theorem A. Section 4 constructs the gauge slices and proves that they miss no vacuum. Section 5 proves Theorem B. Section 6 discusses what remains open. Appendix A lists all F-term equations, Appendix B derives (1), and Appendix C describes the certificates and the data.

2. The quantum mechanics of the D-brane system

2.1. **Fields.** We follow [9, Section 2]. Put $N_1 = N_2 = N_3 = 1$ and $N_4 = N$. For $k \neq \ell$ in $\{1,2,3,4\}$ the chiral field $Z^{(k\ell)}$ is an $N_k \times N_\ell$ complex matrix, an open string from the ℓth to the kth stack, and for $k \in \{1,2,3,4\}$ and $i \in \{1,2,3\}$ the chiral field $\Phi_i^{(k)}$ is an $N_k \times N_k$ matrix describing the motion of the kth stack along the ith two-torus and its dual. The gauge group is

$$G = \mathrm{GL}_1(\mathbb{C})^3 \times \mathrm{GL}_N(\mathbb{C}) \ \supset \ U(1)^3 \times U(N),$$

acting by $Z^{(k\ell)} \mapsto g_k Z^{(k\ell)} g_\ell^{-1}$ and $\Phi_i^{(k)} \mapsto g_k \Phi_i^{(k)} g_k^{-1}$. In the language of representation theory the fields are a representation of the quiver Q with vertices $1,2,3,4$, one arrow $\ell \to k$ for each ordered pair $k \neq \ell$, and three loops at every vertex, of dimension vector $(1,1,1,N)$. We write $\mathcal{R}$ for the vector space of all such representations; it has dimension

$$\dim \mathcal{R} = 6 + 6N + 3 \cdot 3 + 3N^2 = 3N^2 + 6N + 15.$$

2.2. **Superpotential.** For $1 \le k < \ell \le 4$ let $c^{(k\ell)} = c^{(\ell k)}$ be complex parameters, determined by the background metric and B-field on T^6 [8]. The superpotential is $\sqrt{2}\,\mathcal{W}$ with $\mathcal{W} = \mathcal{W}_1 + \mathcal{W}_2 + \mathcal{W}_3 + \mathcal{W}_4$,

$$\mathcal{W}_1 = \sum_{k,\ell,m=1}^{3} \varepsilon^{k\ell m}\, \Phi_m^{(k)} Z^{(k\ell)} Z^{(\ell k)} + \sum_{k=1}^{3} \mathrm{Tr}\Big[\big(\Phi_k^{(k)} I_N - \Phi_k^{(4)}\big) Z^{(4k)} Z^{(k4)} \Big], \tag{2}$$

$$\begin{aligned} \mathcal{W}_2 = {} & Z^{(31)} Z^{(12)} Z^{(23)} + Z^{(13)} Z^{(32)} Z^{(21)} + \mathrm{Tr}\big(Z^{(12)} Z^{(24)} Z^{(41)}\big) + \mathrm{Tr}\big(Z^{(42)} Z^{(21)} Z^{(14)}\big) \\ & - \mathrm{Tr}\big(Z^{(13)} Z^{(34)} Z^{(41)}\big) + \mathrm{Tr}\big(Z^{(31)} Z^{(14)} Z^{(43)}\big) \\ & + \mathrm{Tr}\big(Z^{(34)} Z^{(42)} Z^{(23)}\big) + \mathrm{Tr}\big(Z^{(43)} Z^{(32)} Z^{(24)}\big), \end{aligned} \tag{3}$$

$$\mathcal{W}_3 = \sum_{k,\ell,m=1}^{3} c^{(k\ell)} \varepsilon^{k\ell m}\, \Phi_m^{(k)} + \sum_{k=1}^{3} c^{(k4)} \Big(N\, \Phi_k^{(k)} - \mathrm{Tr}\, \Phi_k^{(4)} \Big), \tag{4}$$

$$\mathcal{W}_4 = -\,\mathrm{Tr}\Big(\Phi_1^{(4)} \big[\Phi_2^{(4)}, \Phi_3^{(4)} \big] \Big). \tag{5}$$

This is [9, Section 2] with the overall factor $\sqrt{2}$ removed, which does not change any of the equations below.[3] The function $\mathcal{W}$ is a polynomial on $\mathcal{R}$, invariant under G.

2.3. **Potential and symmetries.** The potential is $V = V_F + V_D + V_{\mathrm{gauge}}$, where $V_F = \sum_\alpha |\partial \mathcal{W}/\partial \varphi_\alpha|^2$ over all chiral fields φ_α,

$$V_D = \frac{1}{2} \sum_{k=1}^{4} \mathrm{Tr}\Big(\sum_{\ell \neq k} Z^{(k\ell)} Z^{(k\ell)\dagger} - \sum_{\ell \neq k} Z^{(\ell k)\dagger} Z^{(\ell k)} + \sum_{i=1}^{3} \big[\Phi_i^{(k)}, \Phi_i^{(k)\dagger} \big] - c^{(k)} I_{N_k} \Big)^2$$

[3] The coefficient of $\mathcal{W}_2$ was fixed in [9] by a string amplitude, and the minus sign in front of $\mathrm{Tr}(Z^{(13)} Z^{(34)} Z^{(41)})$ was corrected there with respect to [8]. For $N = 1$ neither change affects the count.

with real Fayet–Iliopoulos parameters satisfying $\sum_k c^{(k)} N_k = 0$, and

$$V_{\text{gauge}} = \sum_{k\neq\ell} \sum_i \left\| X_i^{(k)} Z^{(k\ell)} - Z^{(k\ell)} X_i^{(\ell)} \right\|^2 + \sum_k \sum_{i,j} \left\| [X_i^{(k)}, \Phi_j^{(k)}] \right\|^2 + \frac{1}{4} \sum_k \sum_{i,j} \left\| [X_i^{(k)}, X_j^{(k)}] \right\|^2,$$

where $X_i^{(k)}$ are the Hermitian matrices of the spatial components of the gauge fields and $\|A\|^2 = \operatorname{Tr} A^\dagger A$ [9, Section 2].

The potential is invariant under the shifts [9, Section 2]

$$\Phi_m^{(k)} \mapsto \Phi_m^{(k)} + \xi_m \ (k \neq m,\ k, m \leq 3), \qquad \Phi_k^{(k)} \mapsto \Phi_k^{(k)} + \zeta_k, \quad \Phi_k^{(4)} \mapsto \Phi_k^{(4)} + \zeta_k I_N \ (k \leq 3), \tag{6}$$

together with $X_i^{(k)} \mapsto X_i^{(k)} + a_i I_{N_k}$ for all k, with $\xi, \zeta \in \mathbb{C}^3$ and $a \in \mathbb{R}^3$. Indeed, in $\mathcal{W}_1$ and $\mathcal{W}_3$ the fields $\Phi_m^{(k)}$ and $\Phi_m^{(\ell)}$ with $\{k,\ell,m\} = \{1,2,3\}$ enter only through $\varepsilon^{k\ell m}(\Phi_m^{(k)} - \Phi_m^{(\ell)})(Z^{(k\ell)} Z^{(\ell k)} + c^{(k\ell)})$, because $Z^{(k\ell)} Z^{(\ell k)} = Z^{(\ell k)} Z^{(k\ell)}$ for $k, \ell \leq 3$; the field $\Phi_k^{(k)}$ enters only through $\Phi_k^{(k)} I_N - \Phi_k^{(4)}$ and through $c^{(k4)}(N\Phi_k^{(k)} - \operatorname{Tr} \Phi_k^{(4)})$; and the commutators in $\mathcal{W}_4$, V_D and V_{gauge} do not see multiples of the identity. The shifts describe the motion of the centre of mass along T^6, its dual, and $\mathbb{R}^3$.

2.4. **The conjecture.** Let $\widetilde{\mathcal{M}}_c$ be the set of zeros of V modulo $U(1)^3 \times U(N)$, and let $\mathcal{M}_c$ be its quotient by the shifts. The low energy dynamics is that of a supersymmetric particle on $\mathcal{M}_c$, tensored with the free motion along the shift directions, whose quantisation at zero momentum and winding gives a unique state, and with the 28 goldstino zero modes, whose quantisation gives the basic $\mathcal{N} = 8$ supermultiplet [9, Section 2]. If $\mathcal{M}_c$ is a compact Kähler manifold of complex dimension d with Betti numbers b_p, the BPS states with which the supermultiplet is tensored are the harmonic forms on $\mathcal{M}_c$, a p-form carrying $2J_3 = p - d$ [24, 14, 1], so that

$$P(y) = \operatorname{Tr}\, y^{2J_3} = \sum_p b_p\, y^{p-d}, \qquad P(-1) = (-1)^d \chi(\mathcal{M}_c). \tag{7}$$

Conjecture 2.1 (strong zero angular momentum conjecture for the D2-D2-D2-D6 system [8, 9])**.** For generic parameters c, the space $\mathcal{M}_c$ is a finite set of $-\hat{c}(4N)$ points. Equivalently, $P(y) = -\hat{c}(4N)$ is independent of y, and every BPS state is an $SU(2)$ singlet.

The number $-\hat{c}(4N)$ is the index of the $\mathcal{N} = 8$ theory for these charges, and its value is independent of the conjecture: it is computed in a dual description by a D1-D5-Kaluza–Klein monopole system carrying momentum [23], see [9, Section 2], and for these charges there are no multi-centred contributions [20, 21]. The conjecture concerns the angular momentum, which the index does not see.

3. Every vacuum is a simple representation

3.1. **Three consequences of the F-term equations.** The zeros of V_F are the critical points of $\mathcal{W}$. Among the equations $\partial\mathcal{W} = 0$ (listed in full in Appendix A) we use three groups. We write $\{k,\ell,m\} = \{1,2,3\}$ and read the lower indices of $\Phi^{(4)}$ modulo 3.

Lemma 3.1. *At every critical point of $\mathcal{W}$:*

$$Z^{(k\ell)} Z^{(\ell k)} = -c^{(k\ell)} \qquad (1 \leq k < \ell \leq 3), \tag{8}$$

$$Z^{(k4)} Z^{(4k)} = -N\, c^{(k4)} \qquad (1 \leq k \leq 3), \tag{9}$$

$$Z^{(4k)} Z^{(k4)} + \left[\Phi_{k+1}^{(4)}, \Phi_{k+2}^{(4)}\right] = -c^{(k4)}\, I_N \qquad (1 \leq k \leq 3). \tag{10}$$

Proof. We use the convention that the equation $\partial\mathcal{W}/\partial A = 0$ for a matrix field A means $\partial\mathcal{W}/\partial A_{ji} = 0$ for all i, j, so that a term $\mathrm{Tr}(AB)$ contributes the matrix B. The field $\Phi_m^{(k)}$ with $k \neq m$ enters $\mathcal{W}$ only through $\varepsilon^{k\ell m}\Phi_m^{(k)}(Z^{(k\ell)}Z^{(\ell k)} + c^{(k\ell)})$, where ℓ is the third index; since $\varepsilon^{k\ell m} = \pm 1$, its equation is (8). The field $\Phi_k^{(k)}$ enters through $\Phi_k^{(k)}\,\mathrm{Tr}(Z^{(4k)}Z^{(k4)}) + Nc^{(k4)}\Phi_k^{(k)}$, and $\mathrm{Tr}(Z^{(4k)}Z^{(k4)}) = Z^{(k4)}Z^{(4k)}$, which gives (9). The field $\Phi_k^{(4)}$ enters through $-\mathrm{Tr}(\Phi_k^{(4)}Z^{(4k)}Z^{(k4)}) - c^{(k4)}\,\mathrm{Tr}\,\Phi_k^{(4)}$ and through $\mathcal{W}_4$. By cyclicity $\mathcal{W}_4 = -\mathrm{Tr}(\Phi_k^{(4)}[\Phi_{k+1}^{(4)}, \Phi_{k+2}^{(4)}])$ for each k, so the equation for $\Phi_k^{(4)}$ is $-Z^{(4k)}Z^{(k4)} - c^{(k4)}I_N - [\Phi_{k+1}^{(4)}, \Phi_{k+2}^{(4)}] = 0$, which is (10). □

Taking the trace of (10) gives back (9). When all $c^{(k\ell)}$ are nonzero, (8) and (9) show that every $Z^{(k\ell)}$ with $k, \ell \leq 3$ is a nonzero number and every row $Z^{(k4)}$ and column $Z^{(4k)}$ is a nonzero vector. We assume $c^{(k\ell)} \neq 0$ for all $k < \ell$ from now on.

3.2. **Simplicity.** A subrepresentation of a representation $x \in \mathcal{R}$ is a collection of subspaces $S_k \subseteq \mathbb{C}^{N_k}$ with $Z^{(k\ell)}S_\ell \subseteq S_k$ and $\Phi_i^{(k)}S_k \subseteq S_k$ for all k, ℓ, i. The representation is simple if its only subrepresentations are 0 and the whole space.

Proposition 3.2. *Every critical point of $\mathcal{W}$ is a simple representation of Q.*

Proof. Let (S_k) be a subrepresentation of a critical point. Since $Z^{(k\ell)} \neq 0$ for $k \neq \ell \leq 3$ and $N_1 = N_2 = N_3 = 1$, either $S_1 = S_2 = S_3 = \mathbb{C}$ or $S_1 = S_2 = S_3 = 0$: if $S_\ell = \mathbb{C}$ then $S_k \supseteq Z^{(k\ell)}\mathbb{C} = \mathbb{C}$.

Suppose $S_1 = S_2 = S_3 = \mathbb{C}$ and $S_4 \neq \mathbb{C}^N$. Then S_4 contains the three columns $Z^{(4k)} = Z^{(4k)} \cdot 1$ and is invariant under $\Phi_1^{(4)}, \Phi_2^{(4)}, \Phi_3^{(4)}$. Let $U = \mathbb{C}^N/S_4$, of dimension $u \geq 1$, and let ϕ_i be the endomorphisms of U induced by $\Phi_i^{(4)}$. The image of $Z^{(4k)}Z^{(k4)}$ is contained in $\mathbb{C}Z^{(4k)} \subseteq S_4$, so this matrix induces 0 on U, and (10) induces $[\phi_{k+1}, \phi_{k+2}] = -c^{(k4)}I_U$. The trace of a commutator is zero, so $c^{(k4)}u = 0$, a contradiction.

Suppose $S_1 = S_2 = S_3 = 0$ and $S_4 \neq 0$. Then S_4 is invariant under the $\Phi_i^{(4)}$ and $Z^{(k4)}S_4 \subseteq S_k = 0$ for $k = 1, 2, 3$. Restricting (10) to S_4, the term $Z^{(4k)}Z^{(k4)}$ vanishes on S_4 and $[\Phi_{k+1}^{(4)}|_{S_4}, \Phi_{k+2}^{(4)}|_{S_4}] = -c^{(k4)}I_{S_4}$, whose trace gives $c^{(k4)}\dim S_4 = 0$, again a contradiction.

So (S_k) is either 0 or everything. □

Remark 3.3. The proof uses only one of the three equations (10), and only through its trace on a quotient or a subspace. It does not use $\mathcal{W}_2$ at all, so it holds for any coefficients of the cubic couplings of the bifundamentals, and in particular is insensitive to the normalisation and the sign discussed in [9].

3.3. ***D*-terms, gauge fields and the moduli space.** Let $\mathcal{R}^{\mathrm{s}} \subseteq \mathcal{R}$ be the set of simple representations. It is Zariski-open, and by Schur's lemma the stabiliser of a simple representation in G is the group $\mathbb{C}^\times$ of scalars (t, t, t, tI_N), which acts trivially on $\mathcal{R}$. So $\mathrm{P}G = G/\mathbb{C}^\times$ acts freely on $\mathcal{R}^{\mathrm{s}}$. A simple representation is θ-stable in the sense of King for every character θ with $\sum_k \theta_k N_k = 0$, since stability only constrains proper nonzero subrepresentations [17]. By King's construction of moduli of stable representations [17], see also [7], $\mathcal{R}^{\mathrm{s}}/\mathrm{P}G$ is a smooth quasi-projective variety and a geometric quotient, of dimension $\dim\mathcal{R} - \dim\mathrm{P}G = 3N^2 + 6N + 15 - (N^2 + 2) = 2N^2 + 6N + 13$; the quotient map is a principal $\mathrm{P}G$-bundle, locally trivial in the étale topology, by Luna's slice theorem [18].

Proof of Theorem A. (i) The function μ appearing inside V_D, namely $\mu_k = \sum_\ell Z^{(k\ell)}Z^{(k\ell)\dagger} - \sum_\ell Z^{(\ell k)\dagger}Z^{(\ell k)} + \sum_i [\Phi_i^{(k)}, \Phi_i^{(k)\dagger}]$, is a moment map for the action of $U(1)^3 \times U(N)$ on $\mathcal{R}$ with its flat Kähler metric (the loops contribute the commutator terms). King's theorem [17] states that the G-orbit of a representation x meets $\mu^{-1}(c^{(1)}, c^{(2)}, c^{(3)}, c^{(4)} I_N)$ if and only if x is polystable for the character determined by the Fayet–Iliopoulos parameters, and that the intersection is then a single orbit of $U(1)^3 \times U(N)$. A simple representation is stable for every such character. The F-term equations are G-invariant because $\mathcal{W}$ is, so every complex orbit of F-term solutions consists of simple representations and contains exactly one unitary orbit of solutions of $V_D = 0$.

(ii) At a zero of V_{gauge}, for each i the collection $X_i = (X_i^{(1)}, X_i^{(2)}, X_i^{(3)}, X_i^{(4)})$ commutes with every arrow, $X_i^{(k)} Z^{(k\ell)} = Z^{(k\ell)} X_i^{(\ell)}$, and with every loop, $[X_i^{(k)}, \Phi_j^{(k)}] = 0$. So X_i is an endomorphism of the representation, which is simple by Proposition 3.2 and part (i). By Schur's lemma $X_i = a_i \cdot (1,1,1,I_N)$ with $a_i \in \mathbb{C}$, and a_i is real because $X_i^{(1)}$ is Hermitian. This is the last shift in (6). Conversely such X makes V_{gauge} vanish.

(iii) By (i) and (ii), $\widetilde{\mathcal{M}_c}$ is the set $\{d\mathcal{W} = 0\} \subseteq \mathcal{R}^{\mathrm{s}}$ modulo G. Since $\mathcal{W}$ is G-invariant it descends to a regular function f_c on $\mathcal{R}^{\mathrm{s}}/\mathrm{P}G$. Near any point of $\mathcal{R}^{\mathrm{s}}$ the quotient map is locally, in the étale topology, a product of $\mathcal{R}^{\mathrm{s}}/\mathrm{P}G$ with $\mathrm{P}G$, and in such coordinates $\mathcal{W}$ depends only on the first factor; hence the ideal generated by all the partial derivatives of $\mathcal{W}$ is the pull-back of the ideal generated by the partial derivatives of f_c, and $\{d\mathcal{W} = 0\}/G = \mathrm{Crit}(f_c)$, also as schemes. The shifts (6) commute with G, preserve simplicity and $\mathcal{W}$, and act freely, so $\mathcal{M}_c = \mathrm{Crit}(f_c)/\mathbb{C}^6$.

(iv) The map $\sigma(x) = -x$ on all chiral fields commutes with G and with the shifts up to $\xi \mapsto -\xi$, $\zeta \mapsto -\zeta$. Each of $\mathcal{W}_1, \mathcal{W}_2, \mathcal{W}_4$ is cubic and $\mathcal{W}_3$ is linear, so $\mathcal{W}(-x) = -\mathcal{W}(x)$ and σ maps critical points to critical points; V_D and V_{gauge} are even. Hence σ induces an involution of $\mathcal{M}_c$. The function $u = Z^{(12)}Z^{(23)}Z^{(31)}$ is invariant under G and under the shifts, it is nonzero on vacua by (8), and $u(\sigma x) = -u(x)$. So σ has no fixed point on $\mathcal{M}_c$, and if $\mathcal{M}_c$ is finite its points fall into pairs. □

Remark 3.4. When $\mathcal{M}_c$ is finite and reduced, $P(y)$ equals its number of points, which by part (iv) is even. All the numbers in (1) are even, so part (iv) is a consistency check of the conjecture for every N. Part (iv) also pairs the critical values: $f_c(\sigma p) = -f_c(p)$ (Figure 2).

Remark 3.5. The rescaling $x \mapsto \lambda x$ of all chiral fields together with $c \mapsto \lambda^2 c$ multiplies $\mathcal{W}$ by λ^3, since every term of $\mathcal{W}$ is cubic in the fields and the parameters c have weight two. Hence $\mathcal{M}_{\lambda^2 c} \cong \mathcal{M}_c$ for $\lambda \in \mathbb{C}^\times$, and we may always normalise one of the parameters, say $c^{(12)} = 1$.

4. Gauge slices that miss nothing

To count the points of $\mathcal{M}_c$ we fix the gauge and the shifts. The shifts (6) act freely and transitively on the coordinates

$$\Phi_1^{(1)},\ \Phi_2^{(1)},\ \Phi_3^{(1)},\ \Phi_1^{(2)},\ \Phi_2^{(2)},\ \Phi_3^{(3)}, \tag{11}$$

so we set these six numbers to zero. This uses ξ_2, ξ_3 (on $\Phi_2^{(1)}, \Phi_3^{(1)}$), ξ_1 (on $\Phi_1^{(2)}$) and $\zeta_1, \zeta_2, \zeta_3$. The group GL_1^3 modulo scalars acts on $(Z^{(12)}, Z^{(23)})$ by $(g_1 g_2^{-1}, g_2 g_3^{-1})$, freely and transitively on $(\mathbb{C}^\times)^2$, so we set $Z^{(12)} = Z^{(23)} = 1$, which is allowed because both are nonzero at a vacuum. The remaining group is $\mathrm{GL}_N(\mathbb{C})$, acting on the rows $r_k = Z^{(k4)}$ by $r_k \mapsto r_k g^{-1}$ and on the columns and on $\Phi_i^{(4)}$ accordingly.

Definition 4.1. For $N \le 4$ the *standard slice* $\Sigma_N \subseteq \mathcal{R}$ is defined by the vanishing of (11), by $Z^{(12)} = Z^{(23)} = 1$, and by

- $N = 1$: $r_1 = 1$;
- $N = 2$: $r_1 = (1, 0)$, $r_2 = (0, 1)$;
- $N = 3$: r_1, r_2, r_3 equal to the standard basis rows e_1, e_2, e_3;
- $N = 4$: $r_1, r_2, r_3 = e_1, e_2, e_3$ and $r_1\Phi_2^{(4)} = e_4$.

Lemma 4.2. *Let $N \leq 4$ and let $\mathcal{R}_N^\circ \subseteq \mathcal{R}^{\mathrm{s}}$ be the set of simple representations with $Z^{(12)}Z^{(23)} \neq 0$ and*

- *$N = 1$: $r_1 \neq 0$; $N = 2$: r_1, r_2 independent; $N = 3$: r_1, r_2, r_3 independent;*
- *$N = 4$: $r_1, r_2, r_3, r_1\Phi_2^{(4)}$ independent.*

Then every orbit of $G \times \mathbb{C}^6$ in $\mathcal{R}_N^\circ$ meets Σ_N in exactly one point, and $\Sigma_N \cap \{d\mathcal{W} = 0\}$ is identified, as a scheme, with the open subscheme $\mathrm{Crit}(f_c)^\circ/\mathbb{C}^6$ of $\mathcal{M}_c$ lying over $\mathcal{R}_N^\circ$.

Proof. The shifts and the torus were treated above. For the GL_N part, if the rows listed in the definition are independent they form a basis of the space of row vectors when $N \leq 3$ (resp. $N = 4$), and there is a unique $g \in \mathrm{GL}_N$ sending them to the standard basis, because $r_k \mapsto r_k g^{-1}$ and $r_1\Phi_2^{(4)} \mapsto r_1 g^{-1} \cdot g\Phi_2^{(4)} g^{-1} = (r_1\Phi_2^{(4)})g^{-1}$ transform the same way. Thus $\Sigma_N \cap \mathcal{R}_N^\circ$ is a slice for a free action, the map $(\Sigma_N \cap \mathcal{R}_N^\circ) \times G \times \mathbb{C}^6 \to \mathcal{R}_N^\circ$ is an isomorphism onto its image modulo the scalars, and the statement about schemes follows as in the proof of Theorem A(iii). □

The slices are not arbitrary. For $N = 4$ the choice of $r_1\Phi_2^{(4)}$ as the fourth basis row is forced by the following observation, which also shows that the rows of a vacuum never span a Φ-invariant space.

Lemma 4.3. *At every critical point, $r_k\Phi_k^{(4)} \in \mathrm{span}(r_1, r_2, r_3)$ for $k = 1, 2, 3$. Moreover the smallest subspace of row vectors containing r_1, r_2, r_3 and invariant under right multiplication by $\Phi_1^{(4)}, \Phi_2^{(4)}, \Phi_3^{(4)}$ is the whole space.*

Proof. The equation for $Z^{(41)}$ (Appendix A, (12)) reads $r_1(\Phi_1^{(1)} I_N - \Phi_1^{(4)}) + Z^{(12)}r_2 - Z^{(13)}r_3 = 0$, so $r_1\Phi_1^{(4)} = \Phi_1^{(1)}r_1 + Z^{(12)}r_2 - Z^{(13)}r_3$; the equations for $Z^{(42)}$ and $Z^{(43)}$ give the cases $k = 2, 3$ in the same way. For the second statement, let T be that smallest subspace and $T^\perp \subseteq \mathbb{C}^N$ its annihilator. Then $T^\perp$ is invariant under the $\Phi_i^{(4)}$ and killed by r_1, r_2, r_3, so $(0, 0, 0, T^\perp)$ is a subrepresentation, and $T^\perp = 0$ by Proposition 3.2. □

Remark 4.4. The chart $r_1\Phi_1^{(4)} = e_4$, which looks just as natural, contains no vacuum at all: by Lemma 4.3 the vector $r_1\Phi_1^{(4)}$ always lies in the span of r_1, r_2, r_3.

The rows of a vacuum can never be proportional to one another, for any $N \geq 2$.

Lemma 4.5. *Let $N \geq 2$. At every critical point of $\mathcal{W}$ the rows r_1, r_2, r_3 span a space of dimension at least two.*

Proof. By (9) every row is nonzero. Suppose they span a line $L = \mathbb{C}r_1$, so that $r_k = a_k r_1$ with $a_k \neq 0$. By Lemma 4.3, $a_k\, r_1\Phi_k^{(4)} = r_k\Phi_k^{(4)} \in L$ for $k = 1, 2, 3$, so L is invariant under right multiplication by the three matrices $\Phi_k^{(4)}$. By the second statement of Lemma 4.3, L is the whole space of row vectors, so $N = 1$. □

For $N \leq 3$ the complement of $\mathcal{R}_N^\circ$ in the set of vacua consists of the configurations whose rows $r_1, \ldots, r_{\min(N,3)}$ entering Definition 4.1 are dependent. For $N = 1$ it is empty, since $r_1 \neq 0$. For $N = 2$ and $N = 3$ we cover it by finitely many further slices, which we call the *exceptional charts*.

- $N = 2$: r_1 and r_2 are dependent and nonzero, so after a change of basis $r_1 = (1, 0)$ and $r_2 = \lambda(1, 0)$ with $\lambda \in \mathbb{C}$. We impose the shift and torus gauge and these two conditions; λ and r_3 are free variables.
- $N = 3$: by Lemma 4.5 the rows span a plane P. Some two of them, r_a, r_b with $a < b$, form a basis of P; after a change of basis $r_a = e_1$, $r_b = e_2$, and the third row r_c lies in $\operatorname{span}(e_1, e_2)$. The matrices g with $e_1 g = e_1$ and $e_2 g = e_2$ preserve these conditions. By Lemma 4.3 the plane P is not invariant under the $\Phi_j^{(4)}$, and $e_1\Phi_a^{(4)} = r_a\Phi_a^{(4)}$ and $e_2\Phi_b^{(4)} = r_b\Phi_b^{(4)}$ lie in P; hence one of the four vectors $e_1\Phi_j^{(4)}$ $(j \neq a)$ and $e_2\Phi_j^{(4)}$ $(j \neq b)$ has a nonzero third coordinate, and a unique such g makes that vector equal to e_3. This gives four charts for each of the three pairs $(a, b) = (1, 2), (1, 3), (2, 3)$, twelve in all, each with all of the gauge freedom fixed.

Every vacuum outside $\mathcal{R}_N^\circ$ lies in one of these charts. For $N = 2$ we show in Section 5 that the chart contains no solution of the F-term equations; for $N = 3$ we can show this for the twelve charts only modulo primes (Section 5.4).

5. Exact counts

5.1. **Method.** Fix rational parameters c_0. On a slice the equations $d\mathcal{W} = 0$ become a system of polynomial equations with rational coefficients. We first eliminate every variable that occurs linearly with a constant coefficient in one equation, which leaves an equivalent system in fewer variables. For a zero-dimensional ideal $I \subseteq \mathbb{Q}[x_1, \ldots, x_n]$ we use two independent facts.

Upper bound. The dimension $\dim_{\mathbb{Q}} \mathbb{Q}[x]/I$, read off from a Gröbner basis, bounds the number of points of $V(I)$ over $\mathbb{C}$, with equality exactly when I is radical [12, Chapter 5].

Lower bound. A rational univariate representation consists of a linear form t in the variables, a squarefree polynomial $\chi \in \mathbb{Q}[t]$ and rational functions $x_i = \rho_i(t)$, with denominators prime to χ, such that the points of $V(I)$ are $(\rho_1(\theta), \ldots, \rho_n(\theta))$ for the roots θ of χ [19]. Given such data, one checks that every generator of I vanishes at $(\rho_i(t))$ modulo $\chi(t)$, and that χ is squarefree. Then the $\deg \chi$ points are distinct solutions, because their t-coordinates are distinct. This check uses only exact polynomial arithmetic over $\mathbb{Q}$ and does not depend on how the representation was found.

5.2. $N = 1$**.** For $N = 1$ the commutator term $\mathcal{W}_4$ vanishes and Σ_1 covers all vacua. By Remark 3.5 we may set $c^{(12)} = 1$. A Gröbner basis of the slice equations over the field $K = \mathbb{Q}(c^{(13)}, c^{(23)}, c^{(14)}, c^{(24)}, c^{(34)})$, computed in Singular, gives $\dim_K K[x]/I = 12$. Specialising this Gröbner basis is valid outside the zero set of the finitely many denominators and leading coefficients that occur in it and in the standard representations that express its elements through the generators and the generators through it, so for all c outside a proper Zariski-closed subset the slice equations have at most 12 solutions, counted with multiplicity. At $c = c_0$ of Table 1 a Gröbner basis over $\mathbb{Q}$ gives $\dim_{\mathbb{Q}} \mathbb{Q}[x]/I = 12$, and the rational univariate representation has $\deg \chi = 12$ and χ squarefree, so there are twelve distinct solutions, the ideal is radical, and each solution is nondegenerate by the argument of Section 5.6; that argument then gives at least 12 distinct solutions for all c outside a proper Zariski-closed subset. The two bounds together give exactly 12 points, each of multiplicity one. This proves Theorem B(i).[4]

[4]The count 12 was obtained in [8] by computing the Hilbert series of the ideal at chosen values of the parameters. The generic statement over the field of rational functions in the parameters is new, as far as we know.

5.3. $N = 2$. At $c = c_0$ the slice Σ_2 has 27 variables and 39 equations; the linear elimination leaves 15 variables and 21 equations. A Gröbner basis of this ideal over $\mathbb{Q}$, computed with the standard basis algorithm of Singular [16] in exact rational arithmetic, gives $\dim_{\mathbb{Q}} \mathbb{Q}[x]/I = 56$. A rational univariate representation of the 27-variable system with $\deg \chi = 56$ was computed with msolve [3]; χ is squarefree and all 39 equations vanish modulo χ. So $V(I)$ consists of exactly 56 points and I is radical. The exceptional chart $r_2 = \lambda r_1$ gives an ideal whose reduced Gröbner basis over $\mathbb{Q}$ is $\{1\}$, so it contains no solution. Hence $\mathcal{M}_{c_0}$ has exactly 56 points.

5.4. $N = 3$. At $c = c_0$ the slice Σ_3 has 43 variables; after the linear elimination there remain 21 variables and 32 equations. A rational univariate representation with $\deg \chi = 208$ was computed with msolve over $\mathbb{Q}$; χ is squarefree, and every equation vanishes modulo χ, which we checked independently with exact arithmetic in FLINT. So the slice carries at least 208 distinct solutions.

Each of these solutions is nonsingular. Let J be the 32×21 Jacobian matrix of the reduced system, evaluated at $x = \rho(t)$, and let R be a fixed 21×32 matrix of integers. Put $D(t) = \det(RJ)$, reduced modulo χ. If $\gcd(D, \chi) = 1$, then $D(\theta) \neq 0$ at every root θ of χ, so J has rank 21 at every solution. We check this modulo the prime $p = 2^{61} - 1$: there the leading coefficient of χ and the denominators of the representation are units, χ stays squarefree, and the reductions of D and χ are coprime. If $D(\theta) = 0$ for a root θ of χ, then θ and the coefficients of D are integral at every prime above p, and reducing modulo such a prime would give a common root of the reductions of D and χ, which do not have one. So $D(\theta) \neq 0$ for every root.[5] A solution at which the Jacobian has full rank is an isolated point of multiplicity one. This proves Theorem B(iii), using Lemma 4.2 to pass from the slice to $\mathcal{M}_{c_0}$.

We have not been able to show over $\mathbb{Q}$ that there are no further solutions, in the slice or in the twelve exceptional charts. The standard basis algorithm, which settles $N = 2$ in half a minute, did not finish for $N = 3$ within two hours and 7 GB of memory. Modulo primes the computations are fast: the Gröbner basis of the reduced system has $\dim_{\mathbb{F}_p} \mathbb{F}_p[x]/I_p = 208$ for $p = 10^9 + 7$ and $p = 2^{30} + 3$, and every exceptional chart has reduced Gröbner basis $\{1\}$ modulo 65521, $10^9 + 7$ and $2^{30} + 3$. These counts give the following partial upper bound.

Proposition 5.1. *Let $p \in \{10^9 + 7,\ 2^{30} + 3\}$ and let $\mathfrak{p}$ be a prime ideal above p in a number field K. Every solution of the reduced $N = 3$ slice equations at c_0 with coordinates in K that are integral at $\mathfrak{p}$ is one of the 208 solutions above, and no exceptional chart has a solution with coordinates integral at $\mathfrak{p}$.*

Proof. Let $\mathcal{O}$ be the completion of the ring of integers of K at $\mathfrak{p}$, with residue field k, and let $\bar{a}$ denote reduction modulo $\mathfrak{p}$. The equations have coefficients that are integral at p. For a chart, a solution $a \in \mathcal{O}^n$ would reduce to a solution over k of equations that generate the unit ideal over $\mathbb{F}_p$, which is impossible. For the slice, the leading coefficient of χ, the denominators of the ρ_i and the values of the denominator polynomial at the roots of χ are units at p, by the checks above at p, so each of the 208 solutions b lies in $\mathcal{O}'^n$ for a finite extension $\mathcal{O}'$ of $\mathcal{O}$, and their reductions are 208 distinct solutions over $\bar{k}$, because χ stays squarefree modulo p. Since $\dim_{\mathbb{F}_p} \mathbb{F}_p[x]/I_p = 208$, these are all the solutions over $\bar{k}$, each of multiplicity one, so the Jacobian matrix has rank 21 at each of them (which we also checked directly). Now let $a \in \mathcal{O}^n$ be a solution and enlarge $\mathcal{O}$ so that it contains the coordinates of all the b. Then $\bar{a} = \bar{b}$ for one of them. Write f for the vector of equations. Taylor expansion

[5] Working modulo a prime loses nothing here, because the statement to be proved is that something does not vanish. An upper bound on the number of solutions would need the opposite implication, from the absence of solutions modulo p to their absence over $\mathbb{Q}$, which fails in general; this is the difficulty discussed below.

TABLE 1. The rational parameters c_0 used in Theorem B. They were drawn at random from fractions $\pm a/b$ with $1 \le a, b \le 30$ (seed 1 of the archived programs).

$c^{(12)}$	$c^{(13)}$	$c^{(14)}$	$c^{(23)}$	$c^{(24)}$	$c^{(34)}$
$-19/28$	$-9/4$	$5/3$	$21/13$	$-1/4$	$-29/27$

gives $0 = f(a) - f(b) = J(b)(a-b) + q(a-b)$, where every entry of $q(a-b)$ is a combination with coefficients in $\mathcal{O}$ of products of two coordinates of $a-b$. Some 21×21 minor of $J(b)$ is a unit, so $\|a-b\| \le \|a-b\|^2$ in the $\mathfrak{p}$-adic norm. Since $\|a-b\| < 1$, this forces $a = b$. □

So every vacuum at c_0 other than the 208 would have, in the coordinates of the reduced system of the slice or of some chart, coordinates with both $10^9 + 7$ and $2^{30} + 3$ in their denominators. We regard this, together with the agreement of 208 with $-\hat{c}(12)$, as strong evidence that the 208 vacua are all of $\mathcal{M}_{c_0}$, but it is not a proof.

5.5. $N = 4$**: a count modulo a prime.** For $N = 4$ the slice Σ_4 has 63 variables and the linear elimination leaves 32. A Gröbner basis over $\mathbb{Q}$ is beyond our means, but modulo the prime $p = 2^{30} + 3$ msolve computes a Gröbner basis of the reduced system, 4089 polynomials with about 1.4 million terms, and a rational univariate representation over $\mathbb{F}_p$ whose eliminating polynomial has degree 684. We checked with FLINT that this polynomial is squarefree over $\mathbb{F}_p$ and that all 51 equations of the reduced system vanish on the parametrisation modulo the eliminating polynomial. So modulo p the standard slice carries exactly 684 solutions, which is $-\hat{c}(16)$, the number found numerically in [10].

This is evidence, not a proof. The number of solutions over $\mathbb{Q}$ and over $\mathbb{F}_p$ agree for all but finitely many primes, but we cannot certify that p is not one of the exceptions, and we have not examined the configurations outside the standard slice, where $r_1, r_2, r_3, r_1\Phi_2^{(4)}$ are dependent.

5.6. **Nondegeneracy and nearby parameters.**

Proof of Theorem B(ii)–(iv). For $N = 2$ the slice ideal at c_0 is radical and zero-dimensional, and for $N = 3$ the Jacobian matrix has full rank at each of the 208 solutions (Section 5.4). In both cases each of these points is a reduced isolated point of the scheme $\Sigma_N \cap \{d\mathcal{W} = 0\}$, which by Lemma 4.2 is $\mathrm{Crit}(f_{c_0})$ near that point. An isolated point of the zero scheme of the differential df is reduced exactly when the Hessian of f there is nondegenerate, so each of these vacua is a nondegenerate critical point. This gives (ii) and the nondegeneracy in (iii).

For (iv), the critical points of f_c depend algebraically on c, and by the implicit function theorem each nondegenerate critical point of f_{c_0} continues to a unique nondegenerate critical point of f_c for c close to c_0; the continuations of distinct points remain distinct. For the Zariski statement, let $X = \{(c, p) : p \in \Sigma_N,\ d\mathcal{W}_c(p) = 0\}$ and let $Y_1, \dots, Y_s$ be the irreducible components of X that contain the points over c_0. Near each such point X is the graph of a holomorphic map from a neighbourhood of c_0, so each Y_j has dimension 6 and maps dominantly and generically finitely, of some degree d_j, onto the parameter space. There is a nonempty Zariski-open set U over which each Y_j has exactly d_j points and the Y_j do not meet. Since U meets every neighbourhood of c_0, where the fibres contain the n_N continuations, $\sum_j d_j \ge n_N$, and every fibre over U contains at least n_N points of the slice. They are vacua by Lemma 4.2 as long as all $c^{(k\ell)} \neq 0$. □

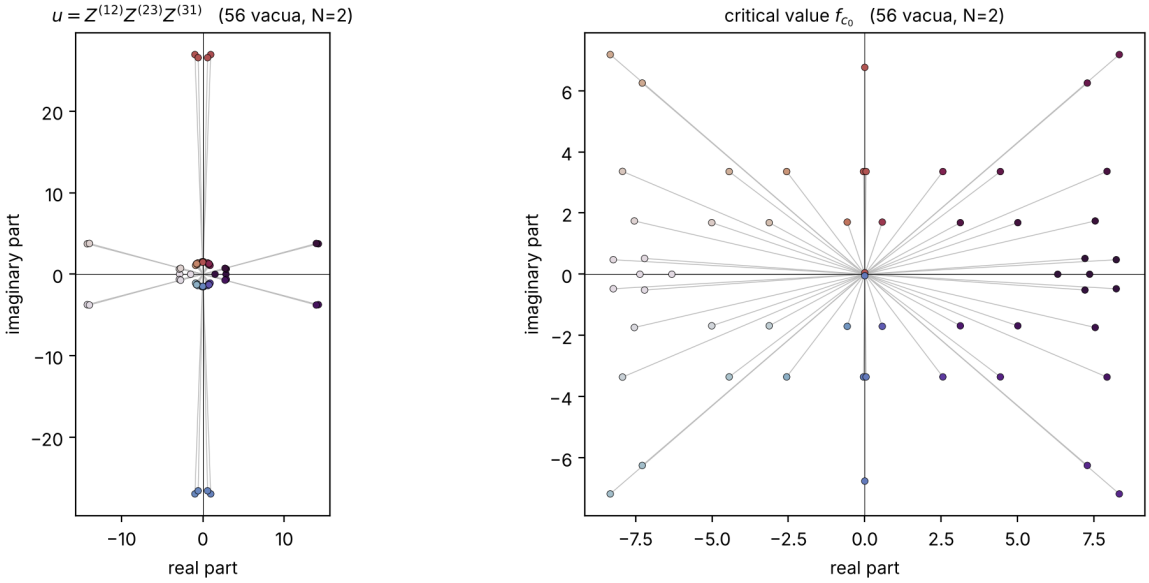


FIGURE 2. The 56 vacua of the $N = 2$ system at $c = c_0$. Left: the gauge-invariant coordinate $u = Z^{(12)}Z^{(23)}Z^{(31)}$. Right: the critical values f_{c_0}. Points joined through the origin are exchanged by the involution σ of Theorem A(iv), which maps $u \mapsto -u$ and $f \mapsto -f$. Because c_0 is real, complex conjugation is a second symmetry, so the points also come in conjugate pairs.

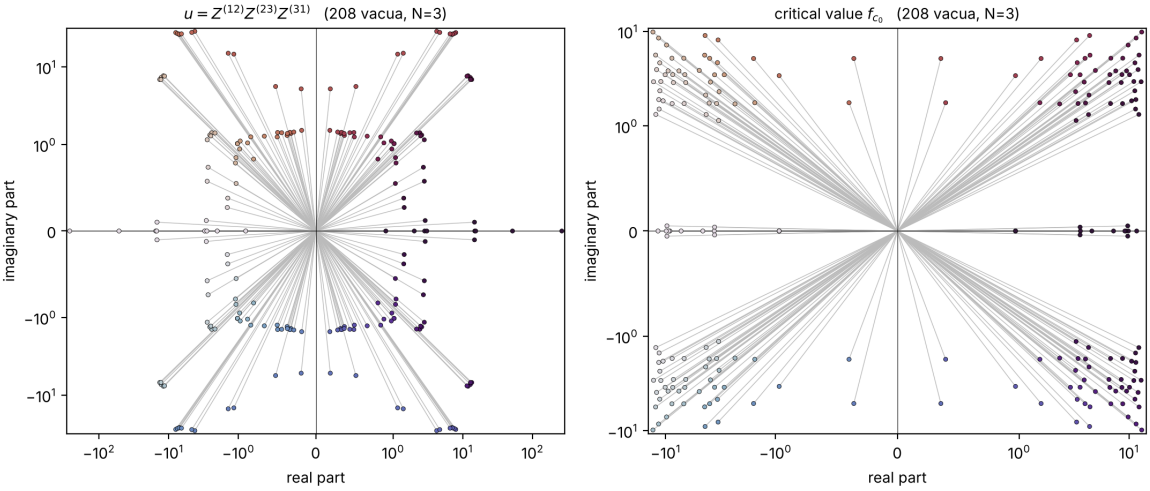


FIGURE 3. The 208 vacua of the $N = 3$ system at $c = c_0$ found in Section 5.4, drawn as in Figure 2 (both axes on a scale that is linear near 0 and logarithmic beyond 1). All 104 pairs $\{p, \sigma p\}$ are visible as segments through the origin, and the 208 critical values are distinct.

6. DISCUSSION

6.1. **What is proved and what is not.** Theorem A holds for every N and every choice of parameters with $c^{(k\ell)} \neq 0$. It reduces the conjecture to the statement that the regular function

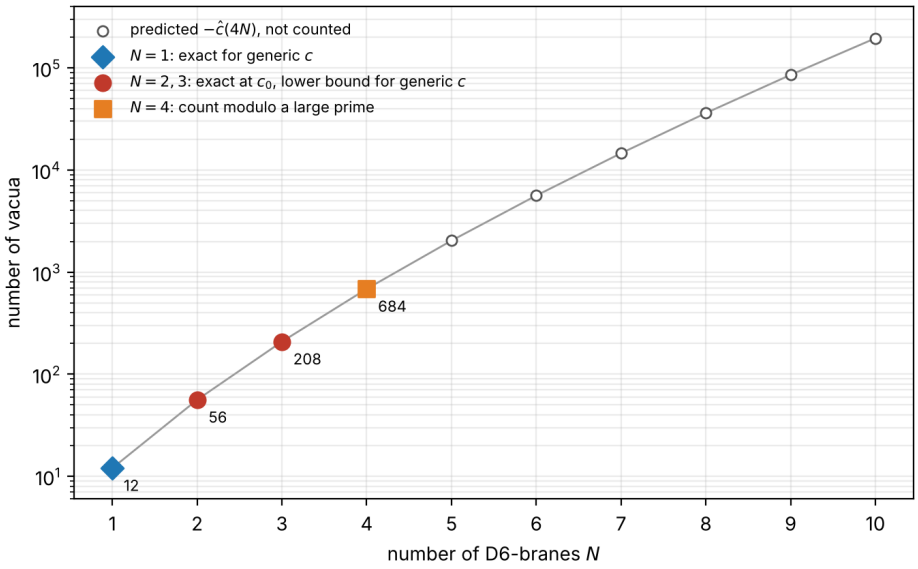


FIGURE 4. The predicted number $-\hat{c}(4N)$ of vacua and the counts of this paper. For $N = 3$ the 208 vacua are certified but it is not proved that there are no others; for $N = 4$ the count is modulo a prime and is not a proof.

f_c on the smooth variety $\mathcal{R}^{\mathrm{s}}/PG$ of dimension $2N^2 + 6N + 13$ has exactly $-\hat{c}(4N)$ critical points modulo the 6-dimensional group of shifts, for generic c, and it shows that the count is even whenever it is finite.

Theorem B gives the count for $N = 1$ at generic parameters and for $N = 2$ at the explicit parameters c_0, the 208 predicted vacua for $N = 3$ at c_0, and lower bounds at generic parameters. Two upper bounds are not proved: for $N = 3$ at c_0, where Proposition 5.1 reduces the question to solutions with large primes in their denominators and an exact Gröbner basis over $\mathbb{Q}$ would settle it, and for $N = 2, 3$ at generic parameters. The reason is that we cannot show that $\mathrm{Crit}(f_c)$ is proper over the parameter space, that is, that no critical point escapes to infinity as c approaches c_0. The usual argument would rescale $x = t\,y$ with $t \to \infty$: an escaping family of critical points of $\mathcal{W}_c$ would approach a nonzero critical point of the cubic part of $\mathcal{W}$, and one would like to show that there is none. But the cubic part does have nonzero critical points, for instance $Z^{(k\ell)} = 0$ for all k, ℓ with commuting $\Phi_1^{(4)}, \Phi_2^{(4)}, \Phi_3^{(4)}$, so this argument does not apply. A proof of the generic upper bound would need either a Gröbner basis over the field of rational functions in the parameters, which is out of reach of present software for $N \geq 2$, or a compactness argument that excludes vacua coming in from infinity.

6.2. **Larger N.** The slice equations grow quickly with N: the slice has $3N^2 + 6N + 15 - (N^2 + 8) = 2N^2 + 6N + 7$ variables, so 27, 43, 63 for $N = 2, 3, 4$. The predicted number of solutions grows like $e^{2\pi\sqrt{N}}$ up to a power of N, the exponential of the entropy $\pi\sqrt{4N}$ of the corresponding black hole (Figure 4). The numerical counts 684, 2032, 5616 of [10, 11] for $N = 4, 5, 6$ match the prediction. By Theorem A each of their solutions of the F-term equations is a simple representation and carries exactly one solution of the D-term equations, so what remains to be proved for these N is the count of critical points alone.

6.3. **Other compactifications.** The same question can be asked for type II string theory on $K3 \times T^2$ and on Calabi–Yau threefolds, whose geometry includes fibrations by polarised $K3$ surfaces and quotients of $K3$ surfaces by involutions [4, 6]. There multi-centred configurations contribute to the index, so a test of the strong conjecture must first separate the single-centred part, and a count of vacua alone does not suffice. The proof of Proposition 3.2 uses only two features of the equations: the bifundamentals between the $U(1)$ nodes are nonzero, and on the $U(N)$ node a commutator of adjoints equals a nonzero multiple of the identity up to a term whose image lies in the span of the incoming arrows and whose kernel contains the common kernel of the outgoing ones. Any quiver theory with these two features has only simple vacua.

6.4. **Two remarks on the structure.** First, the pairing of Theorem A(iv) comes from the oddness of the superpotential, $\mathcal{W}(-x) = -\mathcal{W}(x)$, which holds because every term of $\mathcal{W}$ has odd degree in the fields. It forces every count of isolated vacua to be even, for every N and every c, in agreement with Lemma B.1, which shows that every $\hat{c}(4N)$ is even. Second, Proposition 3.2 does not involve the cubic couplings $\mathcal{W}_2$ of the bifundamentals (Remark 3.3), whereas the counts do: the authors of [9] observe that for $N \geq 2$ the result depends on the coefficient of $\mathcal{W}_2$, which they fix by a string amplitude. All our computations use the superpotential (2)–(5) of [9].

Appendix A. The F-term equations

With the convention of Lemma 3.1, the equations $\partial\mathcal{W}/\partial\varphi = 0$ are the following. For the adjoints they are (8)–(10). For the bifundamentals between the $U(1)$ nodes:

$$
\begin{aligned}
&(\Phi_3^{(1)} - \Phi_3^{(2)})Z^{(21)} + Z^{(23)}Z^{(31)} + Z^{(24)}Z^{(41)} = 0, && (Z^{(12)})\\
&(\Phi_3^{(1)} - \Phi_3^{(2)})Z^{(12)} + Z^{(13)}Z^{(32)} + Z^{(14)}Z^{(42)} = 0, && (Z^{(21)})\\
&(\Phi_2^{(3)} - \Phi_2^{(1)})Z^{(31)} + Z^{(32)}Z^{(21)} - Z^{(34)}Z^{(41)} = 0, && (Z^{(13)})\\
&(\Phi_2^{(3)} - \Phi_2^{(1)})Z^{(13)} + Z^{(12)}Z^{(23)} + Z^{(14)}Z^{(43)} = 0, && (Z^{(31)})\\
&(\Phi_1^{(2)} - \Phi_1^{(3)})Z^{(32)} + Z^{(31)}Z^{(12)} + Z^{(34)}Z^{(42)} = 0, && (Z^{(23)})\\
&(\Phi_1^{(2)} - \Phi_1^{(3)})Z^{(23)} + Z^{(21)}Z^{(13)} + Z^{(24)}Z^{(43)} = 0. && (Z^{(32)})
\end{aligned}
$$

For the bifundamentals of the $U(N)$ node (columns of size N first, then rows):

$$
\begin{aligned}
&(\Phi_1^{(1)} I_N - \Phi_1^{(4)})Z^{(41)} + Z^{(42)}Z^{(21)} + Z^{(43)}Z^{(31)} = 0, && (Z^{(14)})\\
&(\Phi_2^{(2)} I_N - \Phi_2^{(4)})Z^{(42)} + Z^{(41)}Z^{(12)} + Z^{(43)}Z^{(32)} = 0, && (Z^{(24)})\\
&(\Phi_3^{(3)} I_N - \Phi_3^{(4)})Z^{(43)} - Z^{(41)}Z^{(13)} + Z^{(42)}Z^{(23)} = 0, && (Z^{(34)})\\
&Z^{(14)}(\Phi_1^{(1)} I_N - \Phi_1^{(4)}) + Z^{(12)}Z^{(24)} - Z^{(13)}Z^{(34)} = 0, && (Z^{(41)})\\
&Z^{(24)}(\Phi_2^{(2)} I_N - \Phi_2^{(4)}) + Z^{(21)}Z^{(14)} + Z^{(23)}Z^{(34)} = 0, && (Z^{(42)})\\
&Z^{(34)}(\Phi_3^{(3)} I_N - \Phi_3^{(4)}) + Z^{(31)}Z^{(14)} + Z^{(32)}Z^{(24)} = 0. && (Z^{(43)})
\end{aligned}
\tag{12}
$$

Each line is obtained by collecting the terms of $\mathcal{W}$ that contain the given field and deleting that field, keeping the cyclic order: for instance $Z^{(12)}$ occurs in $\varepsilon^{123}\Phi_3^{(1)}Z^{(12)}Z^{(21)}$, $\varepsilon^{213}\Phi_3^{(2)}Z^{(21)}Z^{(12)}$, $Z^{(31)}Z^{(12)}Z^{(23)}$ and $\mathrm{Tr}(Z^{(12)}Z^{(24)}Z^{(41)})$. In total there are $6+6N+9+3N^2$ scalar equations, one for each coordinate of $\mathcal{R}$. The list was checked against a symbolic differentiation of (2)–(5) for $N = 1, 2, 3, 4$ [5].

Appendix B. The predicted number of vacua

The Jacobi theta function is $\vartheta_1(z|\tau) = -iq^{1/8}(y^{1/2}-y^{-1/2})\prod_{n\geq 1}(1-q^n)(1-q^n y)(1-q^n/y)$ with $q = e^{2\pi i\tau}$, $y = e^{2\pi iz}$, and $\eta(\tau) = q^{1/24}\prod_{n\geq 1}(1-q^n)$, whose 24th power is Ramanujan's cusp form $\Delta(\tau) = \sum_n \tau(n)q^n$ (see [2] for the function $\tau(n)$ and $PSL(2,\mathbb{Z})$). Hence

$$-\frac{\vartheta_1(z|\tau)^2}{\eta(\tau)^6} = (y-2+y^{-1})\prod_{n\geq 1}\frac{(1-q^n y)^2(1-q^n/y)^2}{(1-q^n)^4} = \sum_{k\geq 0,\ \ell\in\mathbb{Z}} \hat{c}(4k-\ell^2)\, q^k y^\ell.$$

The coefficient depends only on $4k-\ell^2$, because $-\vartheta_1^2/\eta^6$ is a weak Jacobi form of weight -2 and index 1 [15, 23].[6] Expanding the product to order q^{10} gives, for $N = 0, 1, \ldots, 10$,

$$-\hat{c}(4N) = 2,\ 12,\ 56,\ 208,\ 684,\ 2032,\ 5616,\ 14592,\ 36088,\ 85500,\ 195312.$$

The value $-\hat{c}(0) = 2$ is the coefficient $-(-2)$ of y^0 in $y-2+y^{-1}$.

Lemma B.1. *$\hat{c}(4N)$ is even for every $N \geq 0$.*

Proof. Work with power series in q whose coefficients are Laurent polynomials in y over $\mathbb{Z}/2$. There $(1-q^n y)^2 = 1-q^{2n}y^2$, $(1-q^n/y)^2 = 1-q^{2n}y^{-2}$, $(1-q^n)^4 = 1-q^{4n}$ and $y-2+y^{-1} = y+y^{-1}$. Hence modulo 2 the series $-\vartheta_1^2/\eta^6$ equals $(y+y^{-1})F$, where F involves only even powers of y. Every term of $(y+y^{-1})F$ has an odd power of y, so the coefficient of $q^N y^0$ is even. □

Lemma B.1 and Theorem A(iv) are two independent explanations of the same parity, one on each side of the duality. For the D2-D2-D2-D6 system with N D6-branes the index is $-\hat{c}(4N)$ [9, Section 2], [23].

Appendix C. Certificates and data

The archived programs [5] do the following.

(i) Build $\mathcal{W}$ symbolically, differentiate it, and impose the slice of Definition 4.1 or an exceptional chart, at c_0 of Table 1.
(ii) Eliminate linearly occurring variables and clear denominators.
(iii) For $N \leq 2$, compute $\dim_{\mathbb{Q}} \mathbb{Q}[x]/I$ and the reduced Gröbner bases of the exceptional charts with the standard basis algorithm of Singular over $\mathbb{Q}$, in exact arithmetic (not with modular methods, whose final test in Singular checks only that the input lies in the ideal of the result).
(iv) Compute rational univariate representations with msolve over $\mathbb{Q}$.
(v) Check each representation independently: parse χ and the ρ_i, verify that $\gcd(\chi,\chi') = 1$, and reduce every equation at $x_i = \rho_i(t)$ modulo χ with exact rational arithmetic in FLINT.
(vi) Check that every solution of a representation is nonsingular, by the determinant test of Section 5.4 modulo $2^{61}-1$ (for $N = 1, 2, 3$), and repeat the test modulo 10^9+7 and $2^{30}+3$ for $N = 3$, as used in Proposition 5.1.
(vii) For $N = 3$, compute modulo 65521, 10^9+7 and $2^{30}+3$ the Gröbner bases of the reduced slice system and of the twelve exceptional charts with msolve.
(viii) For $N = 4$, check the representation modulo $p = 2^{30}+3$ with arithmetic in $\mathbb{F}_p[t]$; this part is not a proof.

The certificate files are part of the archive. Floating point numbers appear only in the figures, which locate the roots of χ with certified complex root isolation and evaluate the coordinates.

[6] For a Jacobi form of index 1 the coefficient of $q^k y^\ell$ is unchanged under $\ell \mapsto \ell+2$, $k \mapsto k+\ell+1$, which preserves $4k-\ell^2$, and under $\ell \mapsto -\ell$; both follow from the transformation laws of a Jacobi form [15, Chapter I].

Declarations

Data availability. The programs and the certificate files are archived at [5].

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